

ALI NIKKHAH

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SUMMARY

AI Engineer with 2+ years of experience designing and deploying production agentic systems, LLM-powered backends, and intelligent retrieval infrastructure. At Digikala, led the enterprise AI intelligence layer serving 5M+ daily queries, transitioning from static semantic matching to stateful LangGraph multi-agent pipelines with hybrid RAG (FAISS + Elasticsearch via RRF), HyDE query expansion, and async FastAPI serving backed by vLLM and Triton. At Turquoise Digital, engineered a financial agentic chatbot with real-time voice transcription (Whisper, wav2vec2.0) and dynamic function-calling for secure live portfolio queries. Strong foundation across the full stack: MLOps, distributed LLM inference, and high-concurrency backend systems.

TECHNICAL SKILLS

Languages & Runtimes: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, R, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Time-Series Transformers, ONNX
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, KAG, Graph RAG, FAISS, ChromaDB, Pinecone
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, ESPnet, TensorRT, TensorFlow Lite
Data Engineering: Apache Spark (PySpark), Trino, ClickHouse, Airflow, Kafka, PeerDB, Elasticsearch, Parquet, Delta Lake
MLOps & Serving: Ray, Triton Inference Server, vLLM (PagedAttention), TensorFlow Serving, TensorRT-LLM, MLflow, W&B, DVC
DevOps & Infrastructure: Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Jenkins, GitLab CI
Databases & Monitoring: PostgreSQL, Redis, Prometheus, Grafana, ELK Stack, Metabase, Great Expectations

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 – Aug 2025

- Led the full architectural transition of the enterprise search and knowledge layer from static semantic matching to stateful multi-agent pipelines powered by **LangGraph**; the system now handles over **5 million daily queries** (p99 <200ms) across product discovery, customer support, and internal knowledge retrieval.
- Designed a Modular Control Pipeline (MCP) that cleanly decouples text ingestion, chunking, retrieval, and response synthesis phases, enabling live LLM hot-swapping and zero-downtime vector index migrations without service interruption.
- Built a hybrid retrieval engine combining dense **FAISS** vector search with sparse **Elasticsearch** keyword lookup, fused via Reciprocal Rank Fusion (RRF); implemented **HyDE** query expansion and hierarchical chunking strategies to significantly close the accuracy gap on long-tail and ambiguous product queries (**+23% recall@10**).
- Deployed **Knowledge-Augmented Generation (KAG)** alongside standard RAG to enforce symbolic reasoning constraints on structured product catalog data, substantially reducing factual hallucinations in generated customer-facing answers (**-41% factual error rate**).
- Engineered async **FastAPI** serving layers with multi-tier **Redis** caching and asynchronous request batching; deployed LLMs via **vLLM** with PagedAttention and **Triton Inference Server** to maximize KV-cache utilization and GPU throughput (**3.2x throughput gain, 60% cost reduction per 1K queries**) under burst traffic conditions.

Digikala

Tehran, Iran

AI Engineer

Nov 2024 – Jan 2025

- Prototyped and benchmarked RAG architectures against real production query logs; integrated LangGraph cyclic agent graphs for multi-turn session management, dynamic context re-evaluation, and automated query rewriting loops.
- Evaluated retrieval precision and end-to-end latency across candidate retrieval systems; findings directly shaped the R&D roadmap adopted in the subsequent lead role.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer & Data Scientist

Sep 2025 – Mar 2026

- Engineered an enterprise **LangGraph** agentic chatbot handling customer support ticketing, financial literacy instruction, and live portfolio inquiries; integrated real-time **Whisper** and **wav2vec2.0** voice transcription so call-center audio streams feed the agent directly, eliminating manual ticket creation (**-78% ticket volume, 94% transcription accuracy**).

- Implemented strict data-access validation layers allowing the agent to safely execute compound-profit calculations via dynamic function-calling and retrieve live user equity balances, without ever exposing raw database credentials to the LLM context window.
- Built the AI portfolio allocation system: ingested user risk-tolerance questionnaires, fused historical time-series pricing data with streaming news-sentiment embeddings, and trained time-series transformer models to produce portfolio value forecasts with explicit statistical confidence intervals feeding automated trading allocations (**Sharpe +0.42 vs benchmark**).

HomaCloud

ML Ops Engineer

- Orchestrated distributed model training pipelines on Kubernetes using Ray for workload scheduling and MLflow for experiment tracking and artifact lineage (**15+ models, 50+ experiments tracked**).
- Built model inference APIs in FastAPI backed by TensorFlow Serving; automated full deployment lifecycle with Docker, ArgoCD, and GitHub Actions, cutting release cycles to under 10 minutes (**99.9% deploy success rate**).

Tehran, Iran

Aug 2021 – Feb 2022

SELECTED PROJECTS & RESEARCH

AI Portfolio Allocation & Risk Engine

Core Architect

- Designed an end-to-end asset recommendation system that processes user risk-tolerance questionnaires, fuses historical price time-series with real-time market-sentiment news embeddings, and applies time-series transformer architectures to project future portfolio valuations with statistical confidence intervals.
- Connected model outputs to automated batch trading execution, with explicit guardrails validating order size against computed risk limits before any submission.

Forex Heuristic Market Prediction Engine

Quantitative Developer

- Formalized professional trader chart-reading heuristics — dynamic trendlines, support/resistance zones, and market microstructure signals — into structured feature matrices consumable by deep sequence models.
- Trained sequence-to-sequence networks on these derived representations to detect trend reversals and rank trade execution signals, treating expert visual pattern recognition as a learnable inductive bias.

Turquoise Digital

2025 – 2026

Personal Research

2024

RESEARCH HIGHLIGHTS

University of British Columbia

Research Assistant – Vision-Language Medical Imaging

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** under self-supervised contrastive learning to automate generation of DICOM-compliant radiology reports from raw ultrasound images.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the evaluation benchmark; preprint in preparation.

L3S Research Center (Leibniz University Hannover)

Research Assistant – Video Motion Classification

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders over dense optical flow fields, deliberately suppressing appearance cues so models learn generalizable human-dynamic representations.
- Achieved state-of-the-art accuracy on **UCF-101** and **HMDB-51** benchmarks.

Remote Research Collaboration

Feb 2024 – Nov 2025

Remote Research Collaboration

Apr 2023 – Feb 2024

EDUCATION

Sharif University of Technology

B.Sc. Electrical Engineering – Digital Systems *GPA: 3.65 overall, 3.91 major*

Tehran, Iran

Sept 2020 – Jul 2024